

Tumor tissue microorganisms are closely associated with tumor immune subtypes

Dongmei Ai^{a,*}, Xiaomeng Zhang^a, Qingchuan Zhang^b, Xiaoxin Li^c, Yishu Wang^a, Xiuqin Liu^a, Li C. Xia^{d,**}

^a School of Mathematics and Physics, University of Science and Technology Beijing, Beijing, 100083, China

^b National Engineering Laboratory for Agri-product Quality Traceability, Beijing Technology and Business University, Beijing, 100037, China

^c Beijing VanJee Technology Co., Ltd, Beijing, 100085, China

^d School of Mathematics, South China University of Technology, Guangzhou, 510641, China

ARTICLE INFO

Keywords:

Tumor microenvironment
Tissue microorganisms
RNA-seq
Immune subtype
Gene expression profiling

ABSTRACT

Studies have found that different immune subtypes are present in the same tumor. Different tumor subtypes have different tumor microenvironments (TME). This means that the efficacy of immunotherapy in actual applications will, therefore, have different results. Existing tumor immune subtype studies have mostly focused on immune cells, stromal cells, genes and molecules without considering the presence of microbes. Some studies have shown that microflora can strongly promote many gastrointestinal cancers. The microbiome has, therefore, become an important biomarker and regulatory factor of cancer progression and therapeutic responses. In addition, the presence of microflora can strongly regulate the host immune system, indirectly affecting tumor growth. Taken together, it is important to study the relationships that develop among tumor tissue microorganisms, tumor immune subtype, and the TME. In this study, correlations between microbial abundance, immune cell infiltration, immune gene expression and tumor immune subtype were studied. To accomplish this, tissue microorganisms and immune cell ratios with significant differences between the different cancers were obtained by comparing 203 gastric cancer and intestinal cancer samples. Two immune subtypes of intestinal samples were obtained by K-means clustering algorithm and tissue microorganisms, immune cell ratios and immune-related genes with significant differences between different immune subtypes were screened through Wilcoxon rank sum test. The results showed that *Clostridioides difficile*, *Aspergillus fumigatus*, *Yarrowia lipolytica*, and *Fusarium pseudograminearum* were all closely associated with the identified tumor immune subtypes. Our open-source software is freely available from GitHub at <https://github.com/gutmicrobes/TMM-subtype.git>.

1. Introduction

The tumor microenvironment (TME) is defined as the microenvironment surrounding tumor cells and, thus, an umbrella term for the internal and external environment of tumorigenesis, tumor growth, and metastasis. This includes the microbial population structure [1], immune cell infiltration, and immune gene expression status in tumor tissues. The TME is closely associated with tumorigenesis and tumor progression.

The immune system of a healthy human can eliminate tumor cells that arise from transformation through immune responses. This theory has been used to treat cancer for many decades [2], and it is the basis for

the application of immunotherapy to treat cancer by artificially enhancing or inhibiting immune functions to kill cancer cells. However, researchers have found that immunotherapy has limited efficacy based on colorectal cancer-related experiments [3]. This conclusion is based on differences present among genotypes and phenotypes in the same type of malignancy in different patients. This heterogeneity arises from the presence of different tumor immune subtypes in tumor tissues, making it an important factor affecting antineoplastic treatment. Therefore, different tumor cells will show differences in immune characteristics, ultimately resulting in differences in the clinical efficacy of antineoplastic agents. In 2018, Vestein et al. collected data for different cancers from the TCGA database, which was used for extensive

* Corresponding author.

** Corresponding author.

E-mail addresses: aidongmei@ustb.edu.cn (D. Ai), lcxia@scut.edu.cn (L.C. Xia).

immunogenomic analysis. Finally, the researchers classified over 10,000 tumors comprising 33 diverse cancer types into six immune subtypes based on macrophage or lymphocyte differences in the TMEs and Th1:Th2 cell ratios [4].

Furthermore, existing studies have shown significant differences in microbial population structure [5], immune cell infiltration in tumor immune subtype [6], and immune gene expression [7] in the TMEs of different tumor immune subtypes. The immune microenvironment was found to play an important role in tumorigenesis, tumor invasion, metastasis, and treatment resistance. The identification of a complex tumor immune microenvironment has higher pathological requirements and has gradually changed from the detection of a single marker to the integrated detection of many microenvironmental markers. Therefore, studying the genes associated with different immune subtypes, the microbial population structure of tumor tissues, and immune cell infiltration can provide a more incisive examination of tumor specificity for the purposes of immunotherapy.

A few studies have found immune cell density to be closely associated with tumor immune subtype, and this parameter has been used as a marker for the classification of tumor immune subtypes [8,9]. Furthermore, evidence has shown that the presence of tumor tissue microorganisms can influence cancer treatment through known replication and accumulation [10,11]. Researchers have also found the expression level of immune genes to play a key role in the response to tumor immune subtype-targeting drugs. To study the correlations among immune genes, microbial population structure, as well as immune cell infiltration and tumor immune subtype, we selected the expression data of immune genes closely associated with tumor immune subtype, immune cell ratio data, and relative microbial abundance of tumor tissue as input characteristics to design a model and identify the characteristics associated with the tumor immune subtypes based on these three factors.

2. Materials and methods

We used data from The Cancer Genome Atlas (TCGA) (Retrieved 2009-04-28) [12], including 203 gastric cancer and intestinal cancer primary tumor tissue RNA-seq data and gene expression profile data. [Supplementary Table 1](#) shows the number of samples for each tumor type among the 203 samples.

We correlated tumor immune subtype and tumor microenvironment according to tissue microbial abundance data, immune cell ratio, and immune-related gene expression. To do this, we first extracted reads not previously mapped to the human reference genome from the aforementioned RNA-seq data, and we carried out quality control and assembly. Next, reads were aligned with microbial reference sequences to obtain the relative abundance of microorganisms. The above process was based on the Cancer Genomics Cloud (www.cancer-genomicscloud.org), which is powered by Seven Bridges [13]. We used Docker to push all software to the cloud platform and constructed an RNA-seq data analytic pipeline. [Fig. 1](#) shows RNA-seq data processing and the microbial abundance calculation process.

2.1. Microbial abundance calculation

2.1.1. Screening of unmapped reads

We used SAMtools [14] to remove human genome sequences from BAM format samples and identify all unmapped reads, including paired unmapped reads and single unmapped reads. We then used the command “samtools view -u -f 12 -F 256” to screen out paired unmapped reads, and we converted them to the FASTQ format [15]. When the FASTQ files obtained from screening underwent upstream processing, we added “/1” and “/2” behind the paired read ID and divided them into two FASTQ files containing forward reads and reverse reads so that the reads were compliant with the input format for subsequent software. Finally, we used the command “samtools view -u -f 4 -F 264” to screen out single unmapped reads, which were exported as FASTQ files.

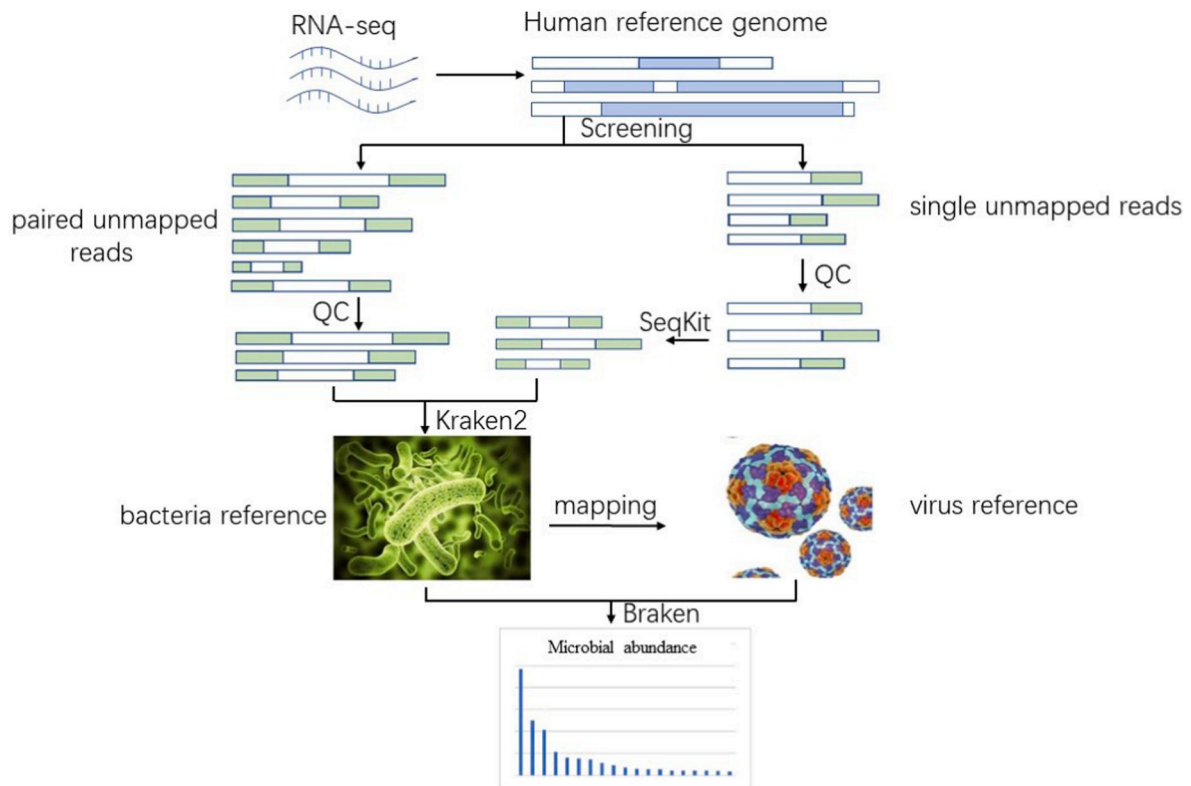


Fig. 1. RNA-seq data processing and microbial abundance calculation procedures.

2.1.2. Quality control

We used Fastp [16] to perform quality control of selected reads not mapped to the human reference genome. We used the command “fastp -q 0 -u 100 -n 10 -l 36 -A -G -M 0 -i” to delete paired-end sequences containing more than 40% reads with a base quality < Q15, sequences with N > 5, sequences with a length < 36, and paired-end adapters. We then used SeqKit [17] to convert the sequence data fastq format to fasta format.

To complete the process, we used the command “-q 0 -u 100 -n 10 -l 36 -A -G -M 0 -i” to perform the aforementioned quality control on single-end sequences. Similarly, SeqKit was used for format conversion of the sequences. To unify the format with paired-end sequences, as the input files for remapping, we again used SeqKit to synthesize the reverse complementary sequences for single unmapped reads, which were used as the paired reads of single unmapped reads. Hence, single unmapped reads were converted to paired-end sequences, which could be combined with the original paired-end sequences.

2.1.3. Estimation of microbial relative abundance and data preprocessing

After preprocessing of sequence data, we used Kraken2 [18] to remap the reads obtained from the above steps to the microbial reference gene bank, and we downloaded and used all intact bacteria, fungi, archaea, and virus genomes from Refseq [19]. A total of 3000 species were used to construct the microbial reference library, and Kraken2 was used to classify the read data. Based on the classification results from Kraken2, we used Bracken [20] to calculate the relative abundances of 2852 bacteria, fungi, and archaea species, and 68 viruses. We used the following criteria for the screening of microbial characteristics.

Command s showed the total number of samples, and c_j represented the number of the j th microorganisms in the sample. We selected microorganisms that satisfied the criteria of $c_j > \frac{1}{2}s$, and, finally, 65 bacteria, fungi, and archaea were selected. Fig. 2 shows the top 10 bacteria, fungi, and archaea according to mean abundance. The same method was used to screen for 68 viruses, and the viruses with a read number of < 6 were removed. Finally, 12 viruses were obtained from screening (Fig. 3).

2.2. Estimation of immune cell ratio in samples based on CIBERSORT

Solid tumor histopathological analysis showed infiltration by innate and adaptive immune cells. Tumor-infiltrating immune cells may be used as drug targets to improve the survival rate of patients; however, overall immune infiltration in tumors varies among patients. Some patients have a high degree of infiltration, while some have a low degree of

infiltration, and others fall somewhere in between. In addition, the distribution of different types of immune cells depend on the chemokines produced by the tumor cells at different sites, vascular and lymphatic system tissues, variable density of innate cells, and intra-tumor adaptive immunity. Significant intra-tumor and inter-tumor differences have been found in the quality and quantity of different active immune cells [21,22].

As noted above, heterogeneity in immune microenvironments will severely affect the clinical response and prognosis of tumor immunotherapy. Precision treatment can only be achieved through a sufficient understanding of the differences among immune cells and other factors in the tumor immune microenvironment. In this section, those differences affecting the composition of infiltrating immune cells in different immune subtypes were analyzed to provide support for the identification of new drug targets.

In a similar study, Thorsson et al. used CIBERSORT [23] to calculate the relative ratios of 22 types of immune cells in 10,000 tumor samples covering 33 cancer types in TCGA, and Supplementary Table 2 names these 22 immune cells. Based on tissue gene expression profile data of the 203 samples, we also employed the CIBERSORT [23] algorithm to calculate the ratios of immune cells in the tumor tissues, and Fig. 4 shows the bar chart of the ratios of 22 types of immune cells.

2.3. Identification of immune-related genes

First, gene expression profile data of 203 samples were downloaded from the TCGA database, and the data consisted of the htseq-count results of 60483 Ensembl Gene IDs. Ensembl Gene IDs consisted of a set of numbers of a gene, including species, type, and version number, while htseq-count was the software used for read counting and counting the number of reads located in certain units of a gene. These units referred to intervals located on the chromosomes [24].

The sample gene expression profile data downloaded contained 19512 genes. To search for genes associated with tumor immune subtypes, we next downloaded a list of 2483 immune-related genes from IMMPort (<https://www.immport.org/>), an open-source database containing all human innate immune-related genes. When we screened for intersections with these immune genes from the gene expression profile data of the 203 samples, we obtained 1355 genes related to innate immunity and created a new gene expression profile.

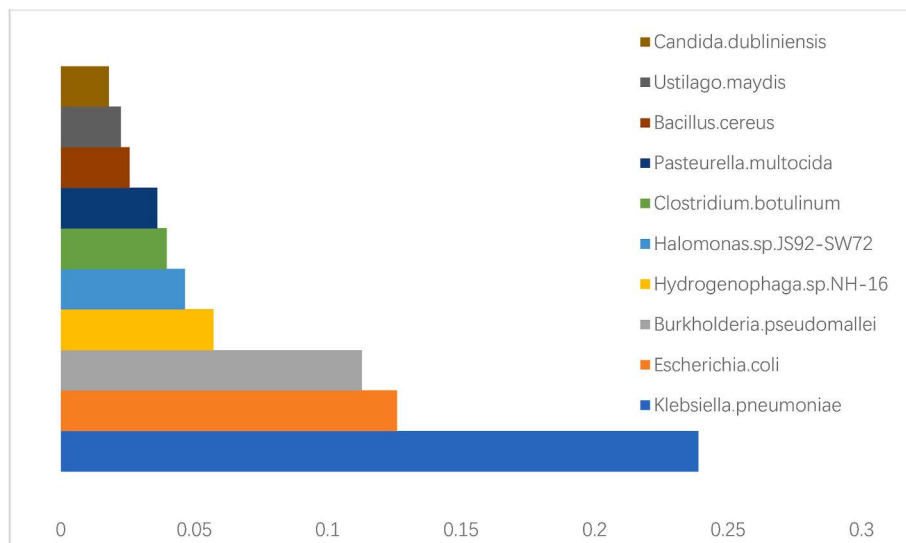


Fig. 2. Top 10 bacteria, fungi, and archaea by mean abundance where the species were arranged by increasing the mean abundance from top to bottom.

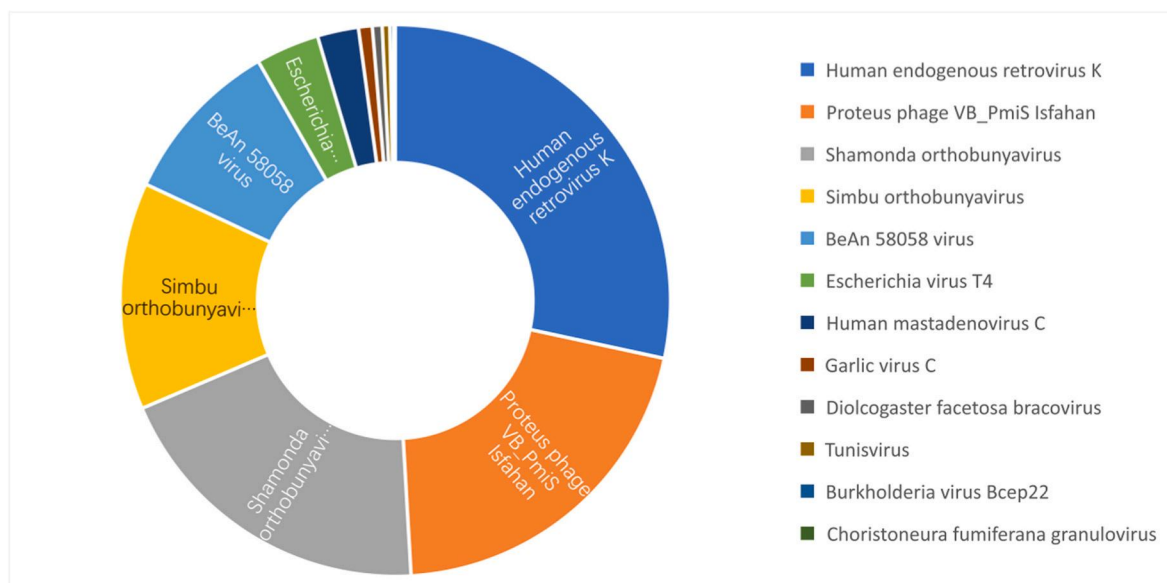


Fig. 3. Mean abundance of 12 viruses arranged clockwise by decreasing mean abundance.

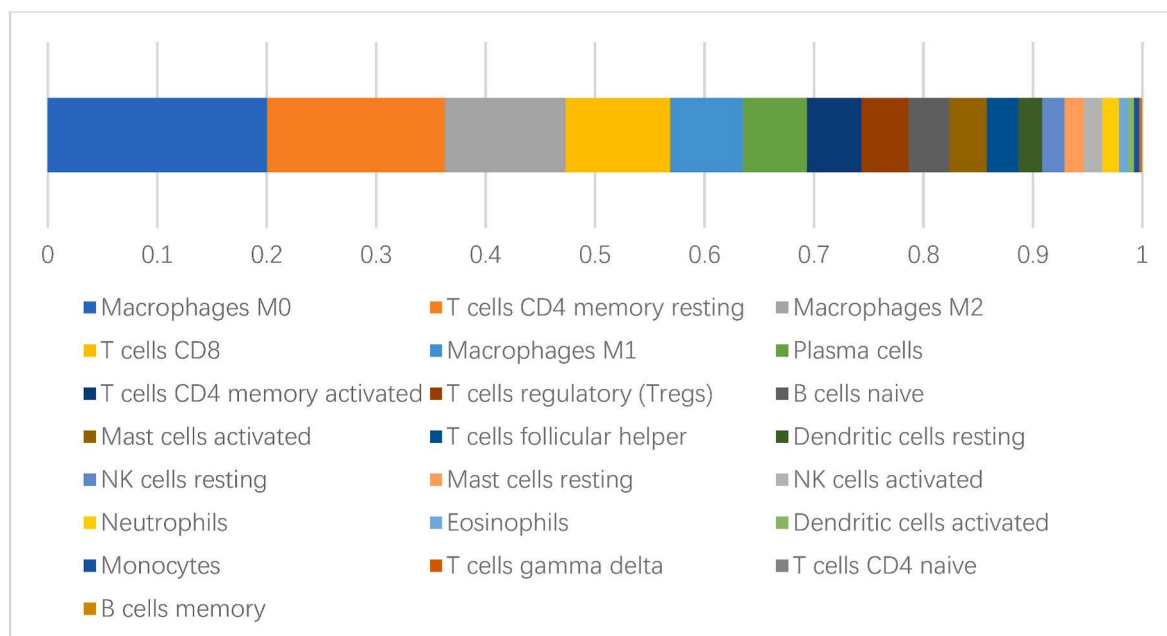


Fig. 4. Average proportion of 22 kinds of immune cells. Different immune cells are represented by different colors. The sum of ratios of 22 types of immune cells was 1.

2.4. Identifying correlations between tumor-associated microorganisms and immune cells

The alignment of microbial abundance data from 153 intestinal cancer patient tissues and 50 gastric cancer patient tissues, immune cell ratio data, and immune gene expression data found large differences in the abundance of 54 microbes, infiltration of 12 immune cells, and the expression characteristics of 734 immune genes between intestinal and gastric cancer patients. In the context of this study, these differential characteristics are considered to play important roles in cancer progression.

Fig. 5 shows the variability among immune cells in different tumors. We found that the levels of Monocytes, T cells regulatory Tregs, Dendritic cells activated, B cells memory, T cells CD4 memory resting, and

Mast cells resting were higher in gastric cancer and that the levels of Macrophages M0, Macrophages M1, Plasma cells, Mast cells activated, T cells follicular helper, and T cells CD4 memory activated were higher in intestinal cancer.

Fig. 6 shows different populations of microorganisms in different tumors. For example, *Aspergillus fumigatus* is a spore-forming mold and one of the most common saprophytic fungi, which has low pathogenicity in otherwise healthy humans, but severe infectivity in immunocompromised patients [25]. *Aspergillus fumigatus* spores are known as conidia, which is often inhaled. According to estimates, conidia are killed by macrophages through oxidase-dependent phagocytosis. However, during steroid treatment, macrophage functionality is completely lost, allowing for conidia germination and hypha formation [26]. Hypha can rapidly grow and destroy mammalian tissues, allowing the fungus to

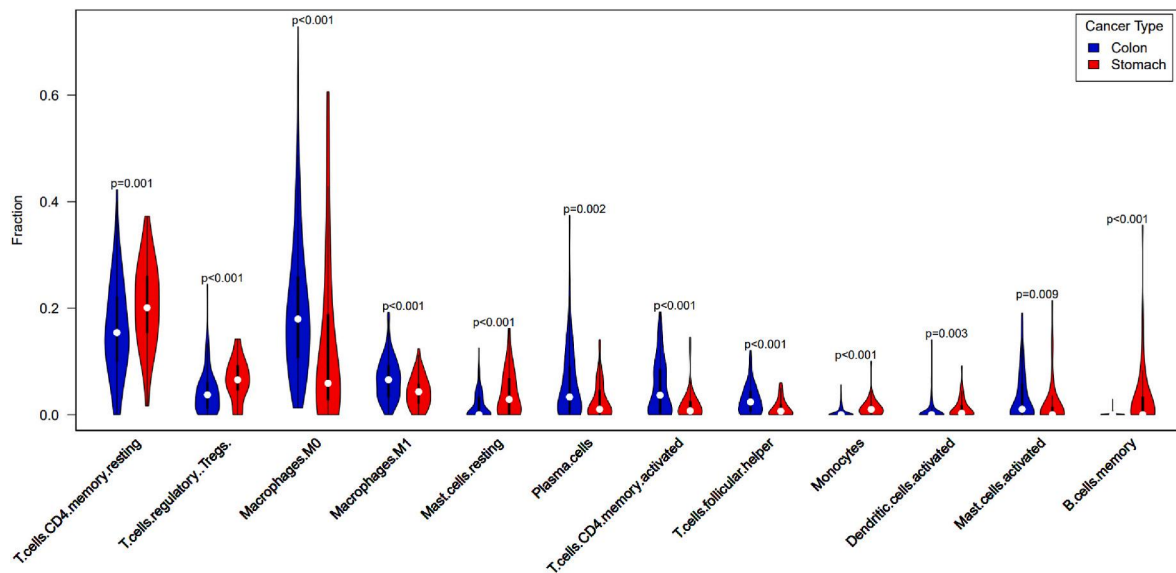


Fig. 5. Immune cells with differences between gastric cancer and colorectal cancer. Blue represents colon cancer, red represents gastric cancer, white dots in the middle are medians, and black rectangles are the upper and lower quartiles.

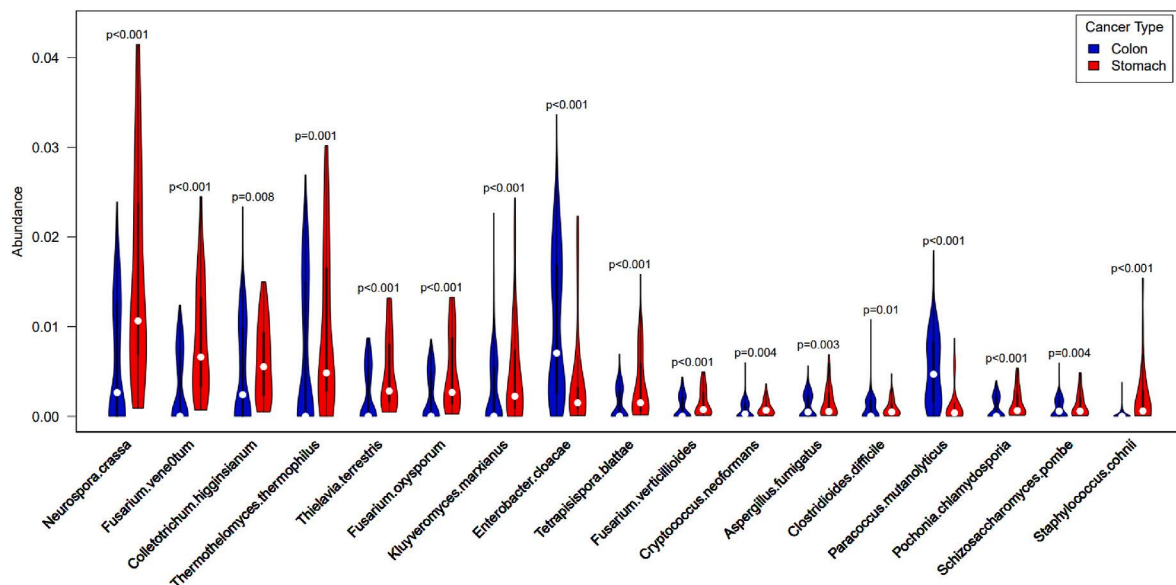


Fig. 6. Microorganisms with differences between gastric cancer and colorectal cancer. Blue represents colon cancer, red represents gastric cancer, white dots in the middle are medians, and black rectangles are the upper and lower quartiles.

further invade tissues and blood vessels, resulting in blood-borne transmission. Although the germination of conidia is mainly controlled by macrophages, neutrophil recruitment and activation will also limit hypha invasion into tissues [26,27].

Clostridioides difficile is a spore-forming anaerobic gut pathogen that can cause severe diarrhea and death [28]. *Clostridioides difficile* infection (CDI) accounts for 15–25% of all opportunistic gastrointestinal infections [29,30], often occurring in inflammatory bowel disease patients, which can cause their clinical presentation to become complex. CDI colonizes the gut, producing two major glycosylated toxins, namely *Clostridioides difficile* toxin A (TcdA) and toxin B (TcdB) [31,32], which can destroy/regulate the host microbiome [33,34]. TcdA and TcdB induce mucosal inflammation and diarrhea to promote pathogenicity [32,35]. This facilitates inflammatory responses beneficial to *Clostridioides difficile* before gradually reaching resident mast cells or recruited immune cells (macrophages and polymorphonuclear cells)

[36,37]. Subsequently, Tcds stimulate colonic epithelial cells and resident or recruited immune cells to secrete proinflammatory cytokines, chemokines, and neutrophil chemokines, amplifying acute inflammatory responses, the main clinical presentation of CDI [32,35]. In addition, TcdA and TcdB can stimulate immune cells to secrete chemokines and cytokines [36,38].

2.5. Colon cancer sample immune subtype clustering

We used the ConsensusClusterPlus software package in R for the unsupervised clustering of immune cell ratio data and immune gene expression data to divide 153 intestinal cancer patients into two different subtypes. In addition, we analyzed how microorganisms, immune cells and immune genes with significant differences between the two subtypes interact with each other, as shown in Fig. 7. Since we used many characteristics, the clustering results were not significant, as

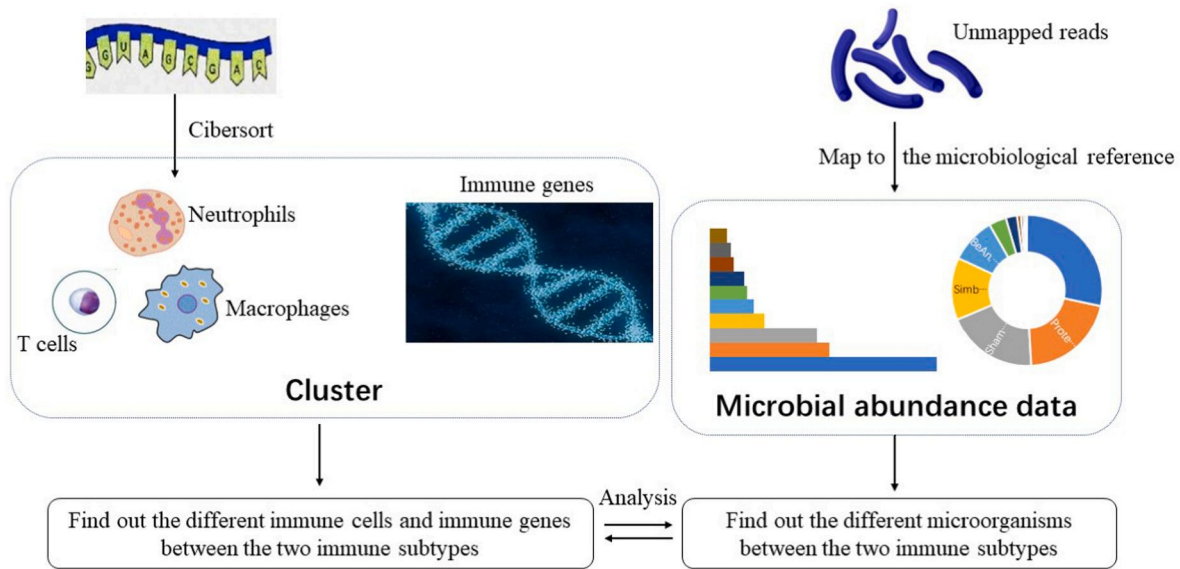


Fig. 7. Cluster flow chart of immune subtypes of colon cancer samples.

shown in Supplementary Fig. 1. Consequently, we used the Lasso algorithm for feature selection by shrinking the coefficients to 0. The minimum value of coefficient β_λ was obtained from the following equation:

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = RSS + \lambda \sum_{j=1}^p |\beta_j|$$

Where $\lambda \geq 0$ is a turning parameter, RSS is residual sum of squares. Lasso used the l_1 penalty, but not the l_2 penalty, and the l_1 norm of coefficient vector β was defined as $\|\beta\|_1 = \sum |\beta_j|$.

The glmnet package in R language was used to carry out the Lasso algorithm. After 1335 immune gene characteristics were inputted, 18 genes were selected, and 22 immune cell characteristics and 18 immune gene characteristics were reclustered. Supplementary Fig. 2 shows the resulting cluster plot.

Figs. 8–10 show the microbial characteristics, immune cell characteristics, and immune gene characteristics with significant differences

among the different subtypes.

The abundance of *Aspergillus fumigatus* was higher in subtype 2 than subtype 1. Because its conidia were phagocytosed and killed by macrophages through oxidase-dependent mechanisms, the densities of Macrophages M1 and Macrophages M2 were higher in subtype 2 than subtype 1. According to one study, the JAK2/STAT3 gene will tend to undergo phosphorylation. JAK/STAT inhibitors can allow *Aspergillus fumigatus* to inhibit STAT3 phosphorylation in neutrophils, impairing its fungicidal activity [39], which was consistent with Fig. 10. This was because *Aspergillus fumigatus* was high in subtype 2, correspondingly causing the JAK2/STAT3 expression level of subtype 2 to be higher than subtype 1. As shown in Fig. 9, the density of Macrophages M1 was higher in subtype 2, while the densities of T cells regulatory Tregs and Plasma cells were higher in subtype 1. In tumor immune typing, the three most widely accepted major tumor immune subtypes are known as immune-infiltrated, immune-excluded, and immune-desert [40,41]. Tregs were enriched in the immune-infiltrated and immune-excluded subtypes [41], whereas macrophages were enriched in the

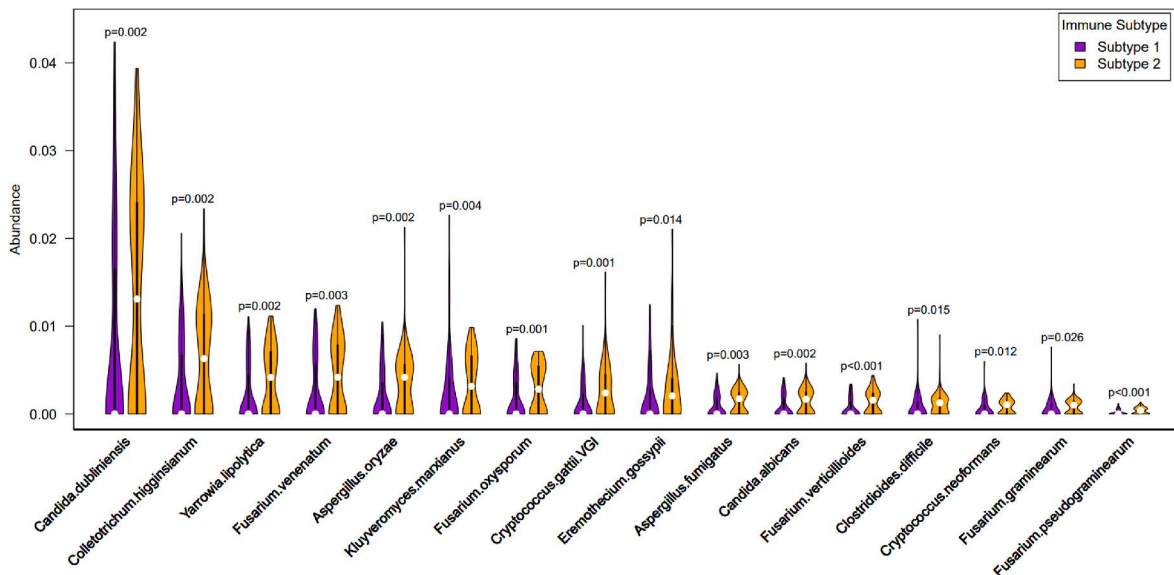


Fig. 8. Microorganisms with abundance differences between the two immune subtypes. Purple represents subtype 1, orange represents subtype 2, white dots in the middle are medians, and black rectangles are the upper and lower quartiles.

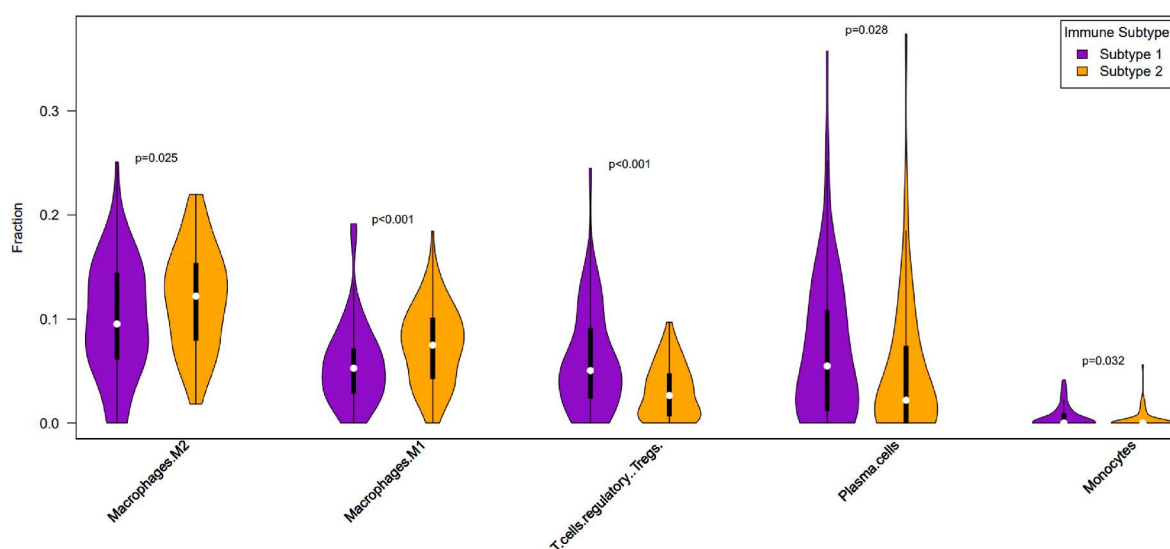


Fig. 9. Immune cells with different ratios between the two immune subtypes. Purple represents subtype 1, orange represents subtype 2, white dots in the middle are medians, and black rectangles are the upper and lower quartiles.

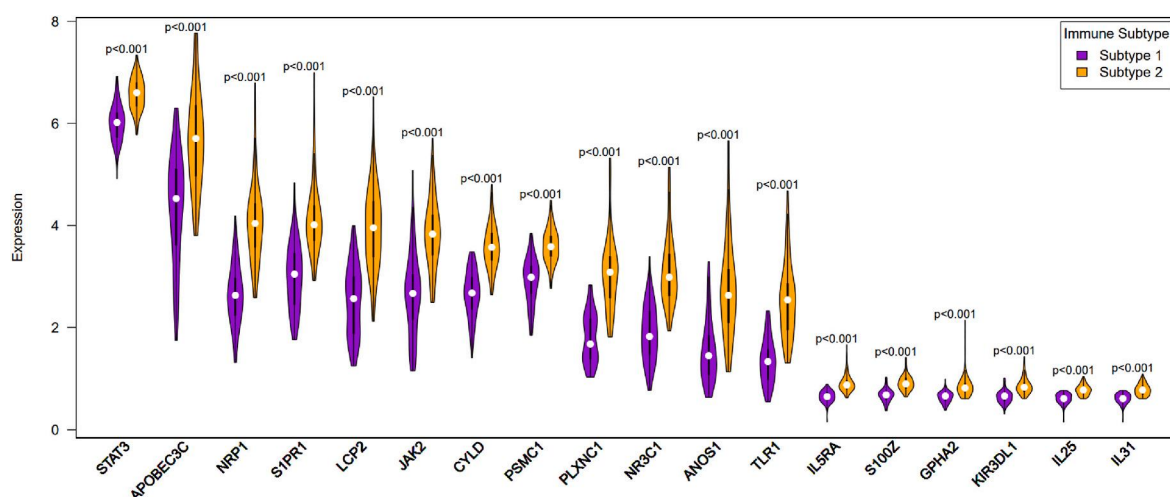


Fig. 10. Immune genes with differences between the two immune subtypes. Purple represents subtype 1, orange represents subtype 2, white dots in the middle are medians, and black rectangles are the upper and lower quartiles.

immune-desert subtype [41]. In a comparative study, we found that our Immune Subtype 1 was closer to the immune-infiltrated and immune-excluded subtypes and that Immune Subtype 2 was closer to the immune-desert subtype.

Yarrowia lipolytica is a lipid-producing yeast that can accumulate large amounts of lipids, up to 50% of the cell's dry weight [42], and it can also produce acetic acid, propionic acid, and butyric acid [43]. Butyric acid can induce neutrophil recruitment [44] and block the proliferation, angiogenesis, and metastasis of colorectal cancer cells by downregulating NRP-1 expression [45]. In addition, butyrate is a ligand of the G-protein coupled receptor that converts CD4+T cells to immunosuppressant Tregs, thereby aiding in the secretion of anti-inflammatory cytokine IL-10 [46]. *Yarrowia lipolytica* is generally recognized as safe by the U.S. Food and Drug Administration, and it can synthesize the serotonin metabolite kynurenic acid (KYNA) [47], which is synthesized through the catalysis of KYNA by kynurenine aminotransferase. Owing to its antioxidant, anti-inflammatory, and anticonvulsant effects, KYNA can promote health [48–50], and according to reports, KYNA offers protective effects for human brains [48] and can inhibit the proliferation of colon cancer cells, such as HT-29

adenocarcinoma cells [49]. A previous study showed that KYNA synthesized by *Y. lipolytica* S12 was present in the biomass and culture medium [47]. Therefore, KYNA-containing yeast biomass could be used as a potential new dietary supplement.

Fusarium graminearum, *Fusarium venenatum*, *Fusarium oxysporum*, *Fusarium verticillioides*, and *Fusarium pseudograminearum* all belong to the genus *Fusarium*, and the main mycotoxins of *Fusarium* are deoxynivalenol (DON) and zearalenone (ZEN). Mycotoxins are mutagenic, teratogenic, and carcinogenic in eukaryotes [51,52]. Mycotoxin-induced immunoregulation can impair macrophage and neutrophil functionality, as well as decrease T lymphocyte and B lymphocyte activity and antibody production, thus affecting innate and adaptive immunity [53]. According to previous reports, the effects of DON on humans is increasing, particularly in inducing gastrointestinal disorders [54,55]. DON can induce anorexia, growth retardation, immune dysregulation, changes in neuroendocrine signaling, as well as induce proinflammatory genes and disrupt growth hormones. At the cellular level, DON can inhibit protein synthesis, disrupt cell signal transduction, differentiation, and proliferation, and even lead to death [56,57]. ZEN has toxic effects on the endocrine and reproductive

systems, resulting in developmental toxicity. As a result of its estrogen activity, it has shown the greatest effects on reproductive organs and their functionality [58,59]. The natural and artificial intake of DON and ZEN will cause these mycotoxins to accumulate in animal and human organs, tissues, and bodily fluids, thus causing related diseases.

3. Results and discussion

The tumor microenvironment plays an important role in tumor pathogenesis. We examined correlations among tumor type, tumor immune subtype, and the tumor microenvironment in three areas: microbial population structure, immune cell ratio, and immune gene expression.

A large body of literature has shown that commensal microorganisms in the human body are closely associated with tumor immune responses. To discover the effector mechanisms between microorganisms and tumor immunity and provide new ideas for improving tumor immunotherapy, we employed Docker to develop a cloud-based platform to create a biological data processing pipeline for RNA-seq samples. First, we extracted reads from RNA-seq data that were not mapped to the human gene sequence reference library. Then, Kraken was used to align unmapped reads to reference libraries containing bacteria, fungi, archaea, and viruses. Bracken was used to estimate the abundance of various microorganisms based on Kraken alignment results. Lastly, the microbial abundance matrix was used to analyze the differences in microbial abundance among the different immune subtypes. *Yarrowia lipolytica* and *Fusarium pseudograminearum* showed significant differences between the immune environments of the two groups ($p_{\text{adj}} < 0.05$). A related study also showed that these microorganisms were closely associated with human immune responses. Therefore, these microorganisms could be used as immune subtype biomarkers and are worthy of further study.

4. Conclusions

We used the CIBERSORT algorithm to estimate the ratios of 22 immune cells in a sample tumor microenvironment. We then selected the aforementioned microbial relative abundance data, immune cell ratio data, and immune gene expression data as input characteristics, followed by using KM clustering and Lasso regression for feature selection and immune subtype clustering. Finally, we obtained two immune subtypes. We calculated the microorganisms with significant differences between the two immune subtypes and finally found that *Aspergillus fumigatus*, *Clostridioides difficile*, *Yarrowia lipolytica*, and *Fusarium pseudograminearum* had closer relationships with immune cells and immune subtypes. After literature validation, we found that these microorganisms affected gastrointestinal cancers and diseases in humans.

Author contributions

D.A. and X.Z. conceived and designed experiments. X.Z. and X.L. performed the experiments and analyzed data. D.A. and L.X. supervised the study and manuscript writing with inputs from D.A., X.Z., Q.Z., X.L. and Y.W.

All authors have read and agreed to the published version of the manuscript.

Funding

This work was supported by grants from the National Natural Science Foundation of China (61873027), and open project of National Engineering Laboratory for Agri-product Quality Traceability (No.AQT-2020-YB6).

Declaration of competing interest

None Declared.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.combiomed.2023.106774>.

References

- [1] T. Jiang, T. Yang, Y. Chen, Y. Miao, Y. Xu, H. Jiang, et al., Emulating interactions between microorganisms and tumor microenvironment to develop cancer theranostics, *Theranostics* 12 (6) (2022) 2833–2859, <https://doi.org/10.7150/thno.70719>.
- [2] P. Sharma, K. Wagner, J.D. Wolchok, J.P. Allison, Novel cancer immunotherapy agents with survival benefit: recent successes and next steps, *Nat. Rev. Cancer* 11 (11) (2011) 805–812, <https://doi.org/10.1038/nrc3153>.
- [3] R. Ma, X. Qu, X. Che, B. Yang, C. Li, K. Hou, et al., Comparative analysis and in vitro experiments of signatures and prognostic value of immune checkpoint genes in colorectal cancer, *OncoTargets Ther.* 14 (2021) 3517–3534, <https://doi.org/10.2147/OTT.S304297>.
- [4] V. Thorsson, D.L. Gibbs, S.D. Brown, D. Wolf, D.S. Bortone, T.H. Ou Yang, et al., The immune landscape of cancer, *Immunity* 48 (4) (2018) 812–830 e814, <https://doi.org/10.1016/j.immuni.2018.03.023>.
- [5] W. Liu, M. Li, X. Zhou, L. Cheng, Research progress in the relationship between microorganisms and tumor immunity, *Chin. J. Microbiol. Immunol.* 38 (10) (2018) 790–795.
- [6] B.S. Nedergaard, M. Ladekar, J.R. Nyengaard, K. Nielsen, A comparative study of the cellular immune response in patients with stage IB cervical squamous cell carcinoma. Low numbers of several immune cell subtypes are strongly associated with relapse of disease within 5 years, *Gynecol. Oncol.* 108 (1) (2008) 106–111, <https://doi.org/10.1016/j.ygyno.2007.08.089>.
- [7] R. Ren, K. Tyryshkin, C.H. Graham, M. Koti, D.R. Siemens, Comprehensive immune transcriptomic analysis in bladder cancer reveals subtype specific immune gene expression patterns of prognostic relevance, *Oncotarget* 8 (41) (2017) 70982–71001, <https://doi.org/10.18632/oncotarget.20237>.
- [8] G. Bindea, B. Mlecnik, W.-H. Fridman, J. Galon, The prognostic impact of anti-cancer immune response: a novel classification of cancer patients, *Semin. Immunopathol.* 33 (4) (2011) 335–340, <https://doi.org/10.1007/s00281-011-0264-x>.
- [9] J. Galon, F. Pages, F.M. Marincola, H.K. Angell, M. Thurin, A. Lugli, et al., Cancer classification using the Immunoscore: a worldwide task force, *J. Transl. Med.* 10 (2012) 205, <https://doi.org/10.1186/1479-5876-10-205>.
- [10] S. Patyar, R. Joshi, D.S. Byrav, A. Prakash, B. Medhi, B.K. Das, Bacteria in cancer therapy: a novel experimental strategy, *J. Biomed. Sci.* 17 (1) (2010) 21, <https://doi.org/10.1186/1423-0127-17-21>.
- [11] C.P.D. Costa, P. Vieira, M. Mendes-Rocha, J. Pereira-Marques, R.M. Ferreira, C. Figueiredo, The tissue-associated microbiota in colorectal cancer: a systematic review, *Cancers (Basel)* 14 (14) (2022), <https://doi.org/10.3390/cancers14143385>.
- [12] Retrieved 2009-04-28. The Cancer Genome Atlas Homepage [Online]. NCI and the NHGRI. [Accessed].
- [13] J.W. Lau, E. Lehnert, A. Sethi, R. Malhotra, G. Kaushik, Z. Onder, et al., The cancer Genomics cloud: collaborative, reproducible, and democratized—a new paradigm in large-scale computational research, *Cancer Res.* 77 (21) (2017) e3, <https://doi.org/10.1158/0008-5472.CAN-17-0387>.
- [14] H. Li, B. Handsaker, A. Wysoker, T. Fennell, J. Ruan, N. Homer, et al., The sequence alignment/map format and SAMtools, *Bioinformatics* 25 (16) (2009) 2078–2079.
- [15] P.J.A. Cock, C.J. Fields, N. Goto, M.L. Heuer, P.M. Rice, The Sanger FASTQ file format for sequences with quality scores, and the Solexa/Illumina FASTQ variants, *Nucleic Acids Res.* 38 (6) (2010) 1767–1771, <https://doi.org/10.1093/nar/gkp1137>.
- [16] S. Chen, Y. Zhou, Y. Chen, J. Gu, fastp: an ultra-fast all-in-one FASTQ preprocessor, *Bioinformatics* 34 (17) (2018) i884–i890, <https://doi.org/10.1093/bioinformatics/bty560>.
- [17] W. Shen, S. Le, Y. Li, F. Hu, SeqKit: a cross-platform and ultrafast toolkit for FASTA/Q file manipulation, *PLoS One* 11 (10) (2016).
- [18] D.E. Wood, S.L. Salzberg, Kraken: ultrafast metagenomic sequence classification using exact alignments, *Genome Biol.* 15 (3) (2014) R46, <https://doi.org/10.1186/gb-2014-15-3-r46>.
- [19] K.D. Pruitt, T. Tatusova, D.R. Maglott, NCBI reference sequences (RefSeq): a curated non-redundant sequence database of genomes, transcripts and proteins, *Nucleic Acids Res.* 35 (Database issue) (2007) D61–D65, <https://doi.org/10.1093/nar/gkl842>.
- [20] J. Lu, F.P. Breitwieser, P. Thielen, S.L. Salzberg, Bracken: estimating species abundance in metagenomics data, *Peer J. Comput. Sci.* 3 (2017) e104.
- [21] W.H. Fridman, F. Pagès, C. Sautès-Fridman, J. Galon, The immune contexture in human tumours: impact on clinical outcome, *Nat. Rev. Cancer* 12 (4) (2012) 298–306, <https://doi.org/10.1038/nrc3245>.
- [22] W.-H. Fridman, M.-C. Dieu-Nosjean, F. Pagès, I. Cremer, D. Damotte, C. Sautès-Fridman, et al., The immune microenvironment of human tumors: general

- significance and clinical impact, *Cancer microenvironment : official journal of the International Cancer Microenvironment Society* 6 (2) (2013) 117–122, <https://doi.org/10.1007/s12307-012-0124-9>.
- [23] A.M. Newman, C.L. Liu, M.R. Green, A.J. Gentles, W. Feng, Y. Xu, et al., Robust enumeration of cell subsets from tissue expression profiles, *Nat. Methods* 12 (5) (2015) 453–457, <https://doi.org/10.1038/nmeth.3337>.
- [24] S. Anders, P.T. Pyl, W. Huber, HTSeq—a Python framework to work with high-throughput sequencing data, *Bioinformatics (Oxford, England)* 31 (2) (2015) 166–169, <https://doi.org/10.1093/bioinformatics/btu638>.
- [25] A. Caroli, D. Fregonese, G. Di Falco, R. D'Inca, *Aspergillus fumigatus* pneumonia during cyclosporine treatment for ulcerative colitis, *Am. J. Gastroenterol.* 95 (10) (2000) 3016–3017, <https://doi.org/10.1111/j.1572-0241.2000.03239.x>.
- [26] A. Schaffner, H. Douglas, A. Braude, Selective protection against conidia by mononuclear and against mycelia by polymorphonuclear phagocytes in resistance to *Aspergillus*. Observations on these two lines of defense in vivo and in vitro with human and mouse phagocytes, *J. Clin. Invest.* 69 (3) (1982) 617–631, <https://doi.org/10.1172/jci110489>.
- [27] A. Schaffner, H. Douglas, A.I. Braude, C.E. Davis, Killing of *Aspergillus* spores depends on the anatomical source of the macrophage, *Infect. Immun.* 42 (3) (1983) 1109–1115, <https://doi.org/10.1128/iai.42.3.1109-1115.1983>.
- [28] B.K. Sandhu, S.M. McBride, *Clostridioides difficile*, *Trends Microbiol.* 26 (12) (2018) 1049–1050, <https://doi.org/10.1016/j.tim.2018.09.004>.
- [29] J. Czepiel, M. Drozd, H. Pituch, E.J. Kuijper, W. Perucki, A. Mielimomka, et al., *Clostridium difficile* infection: review, *Eur. J. Clin. Microbiol. Infect. Dis.* 38 (7) (2019) 1211–1221, <https://doi.org/10.1007/s10096-019-03539-6>.
- [30] J.M. Cho, D.S. Pardi, S. Khanna, Update on treatment of *Clostridioides difficile* infection, *Mayo Clin. Proc.* 95 (4) (2020) 758–769, <https://doi.org/10.1016/j.mayocp.2019.08.006>.
- [31] S. Di Bella, P. Ascenzi, S. Siarakas, N. Petrosillo, A. di Masi, *Clostridium difficile* toxins A and B: insights into pathogenic properties and extraintestinal effects, *Toxins (Basel)* 8 (5) (2016), <https://doi.org/10.3390/toxins8050134>.
- [32] K. Fettucciar, P. Marconi, A. Marchegiani, A. Fruganti, A. Spaterna, G. Bassotti, Invisible steps for a global enemy: molecular strategies adopted by *Clostridioides difficile*, *Therap. Adv. Gastroenterol.* 14 (2021), 17562848211032797, <https://doi.org/10.1177/17562848211032797>.
- [33] C. Rodriguez, B. Taminiau, J. Van Broeck, M. Delmee, G. Daube, *Clostridium difficile* infection and intestinal microbiota interactions, *Microb. Pathog.* 89 (2015) 201–209, <https://doi.org/10.1016/j.micpath.2015.10.018>.
- [34] C. Darkoh, K. Plants-Paris, D. Bishoff, H.L. DuPont, *Clostridium difficile* modulates the gut microbiota by inducing the production of indole, an interkingdom signaling and antimicrobial molecule, *mSystems* 4 (2) (2019), <https://doi.org/10.1128/mSystems.00346-18>.
- [35] R.N. Pruitt, D.B. Lacy, Toward a structural understanding of *Clostridium difficile* toxins A and B, *Front. Cell. Infect. Microbiol.* 2 (2012) 28, <https://doi.org/10.3389/fcimb.2012.00028>.
- [36] X. Sun, T. Savidge, H. Feng, The enterotoxicity of *Clostridium difficile* toxins, *Toxins (Basel)* 2 (7) (2010) 1848–1880, <https://doi.org/10.3390/toxins2071848>.
- [37] R. Chandrasekaran, D.B. Lacy, The role of toxins in *Clostridium difficile* infection, *FEMS Microbiol. Rev.* 41 (6) (2017) 723–750, <https://doi.org/10.1093/femsre/flux048>.
- [38] K. Aktories, C. Schwan, T. Jank, *Clostridium difficile* toxin biology, *Annu. Rev. Microbiol.* 71 (2017) 281–307, <https://doi.org/10.1146/annurev-micro-090816-093458>.
- [39] P.R. Taylor, S. Roy, E.C. Meszaros, Y. Sun, S.J. Howell, C.J. Malemud, et al., JAK/STAT regulation of *Aspergillus fumigatus* corneal infections and IL-6/23-stimulated neutrophil, IL-17, elastase, and MMP9 activity, *J. Leukoc. Biol.* 100 (1) (2016) 213–222, <https://doi.org/10.1189/jlb.4A1015-483R>.
- [40] B. Zhang, Q. Wu, B. Li, D. Wang, L. Wang, Y.L. Zhou, m(6)A regulator-mediated methylation modification patterns and tumor microenvironment infiltration characterization in gastric cancer, *Mol. Cancer* 19 (1) (2020) 53, <https://doi.org/10.1186/s12943-020-01170-0>.
- [41] M. Hornburg, M. Desbois, S. Lu, Y. Guan, A.A. Lo, S. Kaufman, et al., Single-cell dissection of cellular components and interactions shaping the tumor immune phenotypes in ovarian cancer, *Cancer Cell* 39 (7) (2021) 928–944 e926, <https://doi.org/10.1016/j.ccell.2021.04.004>.
- [42] K. Gemperlein, D. Dietrich, M. Kohlstedt, G. Zipf, H.S. Bernauer, C. Wittmann, et al., Polyunsaturated fatty acid production by *Yarrowia lipolytica* employing designed myxobacterial PUFA synthases, *Nat. Commun.* 10 (1) (2019) 4055, <https://doi.org/10.1038/s41467-019-12025-8>.
- [43] A.S. Pereira, S.M. Miranda, M. Lopes, I. Belo, Factors affecting microbial lipids production by *Yarrowia lipolytica* strains from volatile fatty acids: effect of co-substrates, operation mode and oxygen, *J. Biotechnol.* 331 (2021) 37–47, <https://doi.org/10.1016/j.jbiotec.2021.02.014>.
- [44] A. Hayashi, H. Nagao-Kitamoto, S. Kitamoto, C.H. Kim, N. Kamada, The butyrate-producing bacterium *Clostridium butyricum* suppresses *Clostridioides difficile* infection via neutrophil- and antimicrobial cytokine-dependent but GPR43/109a-independent mechanisms, *J. Immunol.* 206 (7) (2021) 1576–1585, <https://doi.org/10.4049/jimmunol.2000353>.
- [45] G. Wang, Y. Yu, Y.Z. Wang, J.J. Wang, R. Guan, Y. Sun, et al., Role of SCFAs in gut microbiome and glycolysis for colorectal cancer therapy, *J. Cell. Physiol.* 234 (10) (2019) 17023–17049, <https://doi.org/10.1002/jcp.28436>.
- [46] P. Goncalves, F. Martel, Butyrate and colorectal cancer: the role of butyrate transport, *Curr. Drug Metabol.* 14 (9) (2013) 994–1008, <https://doi.org/10.2174/1389200211314090006>.
- [47] M. Wrobel-Kwiatkowska, W. Turski, T. Kocki, M. Rakicka-Pustulka, W. Rymowicz, An efficient method for production of kynurenic acid by *Yarrowia lipolytica*, *Yeast* 37 (9–10) (2020) 541–547, <https://doi.org/10.1002/yea.3469>.
- [48] Q. Han, T. Cai, D.A. Tagle, J. Li, Structure, expression, and function of kynurenic aminotransferases in human and rodent brains, *Cell. Mol. Life Sci.* 67 (3) (2010) 353–368, <https://doi.org/10.1007/s00018-009-0166-4>.
- [49] K. Walczak, W.A. Turski, G. Rajtar, Kynurenic acid inhibits colon cancer proliferation in vitro: effects on signaling pathways, *Amino Acids* 46 (10) (2014) 2393–2401, <https://doi.org/10.1007/s00726-014-1790-3>.
- [50] K. Walczak, A. Wnorowski, W.A. Turski, T. Plech, Kynurenic acid and cancer: facts and controversies, *Cell. Mol. Life Sci.* 77 (8) (2020) 1531–1550, <https://doi.org/10.1007/s00018-019-03332-w>.
- [51] N.A. Foroud, F. Eudes, Trichothecenes in cereal grains, *Int. J. Mol. Sci.* 10 (1) (2009) 147–173, <https://doi.org/10.3390/ijms10010147>.
- [52] L. Freire, A.S. Sant'Ana, Modified mycotoxins: an updated review on their formation, detection, occurrence, and toxic effects, *Food Chem. Toxicol.* 111 (2018) 189–205, <https://doi.org/10.1016/j.fct.2017.11.021>.
- [53] I.P. Oswald, D.E. Marin, S. Bouhet, P. Pinton, I. Taranu, F. Accensi, Immunotoxicological risk of mycotoxins for domestic animals, *Food Addit. Contam.* 22 (4) (2005) 354–360, <https://doi.org/10.1080/02652030500058320>.
- [54] J.J. Pestka, A.T. Smolinski, Deoxynivalenol: toxicology and potential effects on humans, *J. Toxicol. Environ. Health B Crit. Rev.* 8 (1) (2005) 39–69, <https://doi.org/10.1080/10937400590889458>.
- [55] (2012). Mycotoxins and Human Health. IARC Sci Publ (158), 87–104.
- [56] H.R. Zhou, Q. Jia, J.J. Pestka, Ribotoxic stress response to the trichothecene deoxynivalenol in the macrophage involves the SRC family kinase Hck, *Toxicol. Sci.* 85 (2) (2005) 916–926, <https://doi.org/10.1093/toxsci/kf1146>.
- [57] J.J. Pestka, Deoxynivalenol: mechanisms of action, human exposure, and toxicological relevance, *Arch. Toxicol.* 84 (9) (2010) 663–679, <https://doi.org/10.1007/s00204-010-0579-8>.
- [58] N. Lorenz, S. Danicke, L. Edler, C. Gottschalk, E. Lassek, D. Marko, et al., A critical evaluation of health risk assessment of modified mycotoxins with a special focus on zearalenone, *Mycotoxin Res.* 35 (1) (2019) 27–46, <https://doi.org/10.1007/s12550-018-0328-z>.
- [59] J. Liu, T. Applegate, Zearalenone (ZEN) in livestock and poultry: dose, toxicokinetics, toxicity and estrogenicity, *Toxins (Basel)* 12 (6) (2020), <https://doi.org/10.3390/toxins12060377>.